



大车运行机构设计计算书 Calculation Sheet for Design of Crane Operation Mechanism

*机构级别: M8

*Mechanism class: M8

T7 L3

大车自重 (除移动载荷外): $DL=47.8t$

Self-weight of crane (except for the moving load): $DL=47.8t$

大车重心高度 (相对轨面): $h_{gc1}=1700mm$ (空载)

Crane's height of center of gravity (relative track surface): $h_{gc1}=1,700mm$ (no-load)

$h_{gc2}=1700mm$ (空载)

$h_{gc2}=1,700mm$ (no-load)

大车运行速度: $v_{yc1}=70m/min$

Crane travelling speed: $v_{yc1}=70m/min$

$1.17m/s$

许用起动时间: $[t_q]=6.0s$

Allowable starting time: $[t_q]=6.0s$

许用制动时间: $[t_z]=6.0s$

Allowable braking time: $[t_z]=6.0s$

许用起、制动加速度: $[a_z]0.30m/s^2$

Allowable starting and braking acceleration: $[a_z]0.30m/s^2$

大车车轮每组车轮数: $n_{ct}=4$

Number of wheels in each set of crane wheels: $n_{ct}=4$

小车轨道高度: $h_b=1336mm$ (相对大车轨顶面)

Trolley track height: $h_b=1,336mm$ (relative to the top of the crane track)

基距: $B=5200mm$

Base: $B=5200mm$

轨距: $S_g=37100mm$

Track gauge: $S_g=37100mm$

最大外伸距 (停车位): $S_t=1850mm$ (小车运行的极限位置)

Maximum outreach distance (parking place): $S_t=1850mm$ (operation limit position of trolley)

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姓名: 彭军

Name: Peng Jun

日期: 2024 年 12 月 16 日

Date: December 16, 2024

职务: 机械工程师

Position: Mechanical Engineer



1. 车轮和轨道

1. Wheel and Track

1.1 选型

1.1 Model Selection

大车车轮数: $n_{nl}=8$

Number of crane wheels: $n_{nl}=8$

选择车轮直径: $D_{yc}=400\text{mm}$

Diameter of selected wheel: $D_{yc}=400\text{mm}$

车轮型号: ZQCL400

Wheel model: ZQCL400

车轮材料: 42CrMo 锻造

Wheel material: 42CrMo forging

$\sigma_b=1080\text{MPa}$ (最小抗拉强度)

$\sigma_b=1080\text{MPa}$ (minimum tensile strength)

车轮轴直径: $d_{lc}=120\text{mm}$

Wheel axle diameter: $d_{lc}=120\text{mm}$

单个车轮重量: $m_{lc}=90\text{kg}$

Single wheel weight: $m_{lc}=90\text{kg}$

大车轨道型号: 43kg/m

Crane track model: 43kg/m

轨道尺寸: $A=130\text{mm}$

Track size: $A=130\text{mm}$

$F=130\text{mm}$

$b=70\text{mm}$

$r=12\text{mm}$

轨道材料: U71Mn

Track material: U71Mn

$\sigma_b=880\text{MPa}$ (最小抗拉强度)

$\sigma_b=880\text{MPa}$ (minimum tensile strength)

2.2 踏面疲劳和静强度校核

2.2 Fatigue and Static Strength Check of Wheel Tread

大车自重引起的轮压: $P_{Lc11}=70.80\text{kN}$

Wheel load caused by the self-weight of crane: $P_{Lc11}=70.80\text{kN}$

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$$P_{Lc12}=48.70\text{kN}$$

由惯性载荷引起的轮压: $P_{Lc31}=0.68\text{kN}$ (满载)

Wheel load caused by inertial load: $P_{Lc31}=0.68\text{kN}$ (full load)

$$P_{Lc32}=0.61\text{kN} \text{ (空载)}$$

$$P_{Lc32}=0.61\text{kN} \text{ (no-load)}$$

移动载荷引起的轮压 (移动载荷在极限位置): $P_{Lc41}=40.53\text{kN}$ (满载)

Wheel load caused by moving load (moving load at the limit position):

$$P_{Lc41}=40.53\text{kN} \text{ (full load)}$$

$$P_{Lc42}=1.46\text{kN} \text{ (空载)}$$

$$P_{Lc42}=1.46\text{kN} \text{ (no-load)}$$

$$P_{Lc5}=26.18\text{kN} \text{ (满载)}$$

$$P_{Lc5}=26.18\text{kN} \text{ (full load)}$$

$$P_{Lc52}=0.94\text{kN} \text{ (空载)}$$

$$P_{Lc52}=0.94\text{kN} \text{ (no-load)}$$

正常工作状态下的最小轮压: $P_{Lcmin}=64.03\text{kN}$

Minimum wheel load in normal working condition: $P_{Lcmin}=64.03\text{kN}$

正常工作状态下的最大轮压: $P_{Lcmax}=135.18\text{kN}$

Maximum wheel load in normal working condition: $P_{Lcmax}=135.18\text{kN}$

$$\text{平均轮压: } P_{Lmean}=111.46\text{kN} \quad \left(P_{Lmean} = \frac{P_{Lmin} + 2P_{Lmax}}{3} \right)$$

$$\text{Average wheel load: } P_{Lmean}=111.46\text{kN} \quad \left(P_{Lmean} = \frac{P_{Lmin} + 2P_{Lmax}}{3} \right)$$

与材料有关的许用线接触应力常数: $k_1=8.5\text{N/mm}^2$ (轨道 $\sigma_b>700$, 车轮 $\sigma_b>1000$)

Allowable line contact stress constant related to the material: $k_1=8.5\text{N/mm}^2$ (track: $\sigma_b>700$, wheel: $\sigma_b>1,000$)

车轮与轨道有效接触宽度: $l=46.0\text{mm}$

Effective contact width between the wheel and the track: $l=46.0\text{mm}$

车轮转速: $n_{yc}=55.70\text{r/min}$

Wheel speed: $n_{yc}=55.70\text{r/min}$

转速系数: $C_1=0.92$ (根据车轮直径和运行速度)

Speed coefficient: $C_1=0.92$ (according to the wheel diameter and speed)

工作级别系数: $C_2=0.80$ (机构级别 M8)

Working class coefficient: $C_2=0.80$ (mechanism class: M8)

疲劳强度校核公式: $P_{Lmean} \leq k_1 D_{yt} l C_1 C_2$

Fatigue strength check formula: $P_{Lmean} \leq k_1 D_{yt} l C_1 C_2$

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$k_1 D_{yt} C_1 C_2 = 115.11 \text{ kN}$ 车轮满足疲劳强度要求!
 $k_1 D_{yt} C_1 C_2 = 115.11 \text{ kN}$ The wheels meet the fatigue strength requirements!

静强度校核: $P_{Lmax} \leq 1.9 k_1 D_{yt} l$
Static strength check: $P_{Lmax} \leq 1.9 k_1 D_{yt} l$

$1.9 k_1 D_{yt} l = 251.71 \text{ kN}$ 车轮满足静强度要求!
 $1.9 k_1 D_{yt} l = 251.71 \text{ kN}$ The wheel meets the static strength requirements!

2. 轴承 2. Bearing

调心滚子轴承: 22320CC/W33
Self-aligning roller bearing: 22320CC/W33

重量: 12.80kg
Weight: 12.80kg

轴承支承距离: $L_Z = 148 \text{ mm}$ (车轮中心到轴承中心的距离)
Bearing support distance: $L_Z = 148 \text{ mm}$ (distance from the wheel center to the bearing center)

轴承外径: $D = 215$
Outer diameter of bearing: $D = 215$

轴承内径: $d_Z = 100 \text{ mm}$
Inner diameter of bearing: $d_Z = 100 \text{ mm}$

宽度: $B = 73 \text{ mm}$
Width: $B = 73 \text{ mm}$

$$L_{10h} = \frac{10^6}{60 n_{ye}} \left(\frac{f_t \times C_r}{P} \right)^\epsilon, P = f_d (X F_r + Y F_a)$$

$C_{r0} = 832 \text{ kN}$

额定寿命: $[L_{10h}] = 25000$ 小时
Rated service life: $[L_{10h}] = 25,000 \text{ h}$

温度系数: $f_t = 1.00$ (工作温度 $< 120^\circ \text{C}$)
Temperature coefficient: $f_t = 1.00$ (operating temperature: $< 120^\circ \text{C}$)

寿命系数: $\epsilon = 3.33$ (滚子轴承)
Service life coefficient: $\epsilon = 3.33$ (roller bearing)

冲击载荷系数: $f_d = 1.20$ (中等冲击力)
Impact load coefficient: $f_d = 1.20$ (medium impact force)

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轴向载荷系数: $\gamma=0.10$

Axial load coefficient: $\gamma=0.10$

轴向载荷: $F_{a1}=0.00\text{kN}$

Axial load: $F_{a1}=0.00\text{kN}$

$$F_{a2}=11.20\text{kN} \quad F_a=\gamma P_L$$

$$\text{径向载荷: } F_{r1}=71.15\text{kN} \quad \left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2} \right)$$

$$\text{Radial load: } F_{r1}=71.15\text{kN} \quad \left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2} \right)$$

$$F_{r2}=40.87\text{kN}$$

判断系数: $e=0.30$

Judgment coefficient: $e=0.30$

径向动载系数: $X=1.001.00$

Radial dynamic load coefficient: $X=1.001.00$

轴向动载系数: $Y=2.302.30$

Axial dynamic load coefficient: $Y=2.302.30$

当量动载荷: $P_1=85.38\text{kN}$

Equivalent dynamic load: $P_1=85.38\text{kN}$

$$P_2=79.96\text{kN}$$

$$P=83.57\text{kN}$$

额定动载荷: $C_r=618.00\text{kN}$

Rated dynamic load: $C_r=618.00\text{kN}$

实际轴承寿命: $L_{10h}=235702$ 小时 轴承满足寿命要求!

Actual service life of the bearing: $L_{10h}=235,702\text{h}$ The bearing meets the service life requirements!

3.电动机

3. Motor

3.1.载荷计算

3.1 Load Calculation

自重载荷: $P_{Gc1}=766.0\text{kN}$ (满载)

Self-weight load: $P_{Gc1}=766.0\text{kN}$ (full load)

$$P_{Gc2}=664.0\text{kN} \text{ (空载)}$$

$$P_{Gc2}=664.0\text{kN} \text{ (no-load)}$$

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大车水平惯性载荷: $P_{Ac1}=23.0\text{kN}$ (满载)

Horizontal inertial load of the crane: $P_{Ac1}=23.0\text{kN}$ (full load)

$P_{Ac2}=19.9\text{kN}$ (空载)

$P_{Ac2}=19.9\text{kN}$ (no-load)

车轮轴承摩擦系数: $\mu_c=0.015$ (滚动轴承)

Friction coefficient of wheel bearing: $\mu_c=0.015$ (roller bearing)

滚动摩擦系数: $k_c=0.60$ (轨道为钢质、圆顶)

Coefficient of rolling friction: $k_c=0.60$ (the track is made of steel, with round top)

附加摩擦阻力系数: $C_f=1.50$ (有轮缘, 分别驱动)

Additional coefficient of frictional resistance: $C_f=1.50$ (with wheel rim, driven respectively)

$$P_m = mg \frac{\mu_c d + 2k_c}{D_{yt}} C_f$$

运行摩擦阻力:

$$P_m = mg \frac{\mu_c d + 2k_c}{D_{yt}} C_f$$

Operating friction resistance:

$P_{mc1}=8.62\text{kN}$ (满载)

$P_{mc1}=8.62\text{kN}$ (full load)

$P_{mc2}=7.47\text{kN}$ (空载)

$P_{mc2}=7.47\text{kN}$ (no-load)

运行坡度阻力: $P_p = mg \cdot \sin \alpha = mg \cdot m_p$

Operating slope resistance: $P_p = mg \cdot \sin \alpha = mg \cdot m_p$

坡道阻力系数: $m_p=0.002$

Slope resistance coefficient: $m_p=0.002$

$P_{pc1}=1.53\text{kN}$ (满载)

$P_{pc1}=1.53\text{kN}$ (full load)

$P_{pc2}=1.33\text{kN}$ (满空载)

$P_{pc2}=1.33\text{kN}$ (full load/no-load)

大车运行静阻力: $P_j=P_m+P_p$

Static resistance of crane travelling: $P_j=P_m+P_p$

$P_{jc1}=10.15\text{kN}$ (满载)

$P_{jc1}=10.15\text{kN}$ (full load)

$P_{jc2}=8.80\text{kN}$ (空载)

$P_{jc2}=8.80\text{kN}$ (no-load)

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3.2.使用系数

3.2 Service Coefficient

工作机最低工况系数: $f_{Bmin}=1.40$ (运行机构, 8h/天, $Z=600$ 次)

Minimum working condition coefficient of working machine: $f_{Bmin}=1.40$
(operation mechanism, 8h/day, $Z=600$ times)

环境温度决定的服务系数: $f_{B1}=1.30$ (50°C)

Service coefficient determined by the ambient temperature: $f_{B1}=1.30$ (50°C)

起动系数: $f_{B1}=0.93$ (负载持续率 40%)

Starting coefficient: $f_{B1}=0.93$ (continuous load rate: 40%)

使用系数: $f=f_{Bmin}f_{B1}f_{B2}=1.69$

Service coefficient: $f=f_{Bmin}f_{B1}f_{B2}=1.69$

减速机使用系数 $f_B=1.7$ 减速器选择满足要求!

Service coefficient of reducer: $f_B=1.7$ The selection of the reducer meets the requirements!

3.3.电动机选型

3.3 Motor Selection

电动机台数: $m=4$ (每个台车一套驱动机构, 一驱动轮, 一从动轮)

Number of motors: $m=4$ (each trolley has a set of driving mechanism, one driving wheel and one driven wheel)

运行机构总效率: $\eta_{cz}=0.95$

Overall efficiency of the operating mechanism: $\eta_{cz}=0.95$

机构静功率:

Static power of the mechanism:

$$P_N = \frac{P_{jc} \cdot v_{yc}}{1000 \eta_{cz} \cdot m} = 3.12 \text{ kW}$$

考虑 1.6 倍增大系数 4.99kW

Considering the enhancement coefficient for 1.6 times: 4.99kW

型号: SK4282ABGH

Model: SK4282ABGH

额定功率: $P_n=5.5\text{kW}$

Rated power: $P_n=5.5\text{kW}$

输出转速: $n_n=1450\text{r/min}$

Output speed: $n_n=1450\text{r/min}$

输出轴的转矩: $M_2=941.0\text{N.m}$

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Torque of the output shaft: $M_2=941.0\text{N}\cdot\text{m}$

机构的理论传动比: $i_0=26.03$ ($i_0=n_n/n_{yt}$)

Theoretical transmission ratio of the mechanism: $i_0=26.03$ ($i_0=n_n/n_{yt}$)

实际传动比: $i=26.25$

Actual transmission ratio: $i=26.25$

平均起动转矩倍数: $\lambda_{As}=2.20$

Average starting torque multiple: $\lambda_{As}=2.20$

使用系数: $f_B=1.7$

Service coefficient: $f_B=1.7$

自重: $m_1=69\text{kg}$

Self-weight: $m_1=69\text{kg}$

转动惯量: $J_1=0.025\text{kg}\cdot\text{m}^2$

Rotational inertia: $J_1=0.025\text{kg}\cdot\text{m}^2$

3.4.减速器输出转矩的校核

3.4 Check for Output Torque of the Reducer

运行阻力:

Operating resistance:

$F_{\max}=P_{m1}+P_{p1}+P_{w1}=10.15\text{kN}$

作用在减速器轴上的转矩:

Torque acting on the reducer shaft:

$$T_{\max} = \frac{F_{\max} D_{yc}}{2m} = 507 \quad \text{N}\cdot\text{m}$$

3.5.实际电动机转速

3.5 Actual Motor Speed

转速校核公式: $n_N = \frac{v_{yt} i}{\pi D_{yt}}$

$$n_N = \frac{v_{yt} i}{\pi D_{yt}}$$

Speed check formula:

电动机实际转速: $n_{yc}=1462\text{m}/\text{min}$ (满载)

Actual motor speed: $n_{yc}=1,462\text{m}/\text{min}$ (full load)

3.6.过载校核

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3.6 Overload Check

$$P_n \geq \frac{1}{m\lambda_m} \left\{ [P_m + P_p + P_{wII}] \frac{v_{yc}}{1000\eta_{cz}} + \frac{\sum J \cdot n^2}{91200t_q} \right\}$$

$$\frac{1}{m\lambda_m} \left\{ [P_m + P_p + P_{wII}] \frac{v_{yc}}{1000\eta_{cz}} + \frac{\sum J \cdot n^2}{91200t_q} \right\} = 4.6 \quad \text{kW}$$

3.7.高速轴制动器（电动机自带）

3.7 High-speed Axle Brake (attached with the motor)

$$M_Z = \frac{1}{m} \left\{ \frac{(P_{wI} + P_p - P_m) D_{yc} \cdot \eta}{2i} + \frac{n \sum J}{9.55t_z} \right\}$$

折算到电机轴上的总转动惯量：

$$\Sigma J = 4.34 \text{kg.m}^2 \quad \sum J' = 1.15m(J_1 + J_2) + \frac{(Q + TL + DL)D_0^2 \eta}{4i^2}$$

Total rotational inertia converted to the motor shaft:

$$\Sigma J = 4.34 \text{kg.m}^2 \quad \sum J' = 1.15m(J_1 + J_2) + \frac{(Q + TL + DL)D_0^2 \eta}{4i^2}$$

制动器的制动力矩： $M_Z = 14.9 \text{N.m}$

Braking torque of the brake: $M_Z = 14.9 \text{N.m}$

最大制动力矩： $M_{Z\max} = 60 \text{N.m}$

Maximum braking torque: $M_{Z\max} = 60 \text{N.m}$

个数： $m_z = 4$ 台

Qty.: $m_2 = 4$ sets

制动器选择满足要求！

The selection of the brake meets the requirements!

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4. 机构起动时间和加速度的验算

4. Checking Calculation for Starting Time and Acceleration of the Mechanism

4.1. 起动时间

4.1 Starting Time

$$t_q = \frac{n \sum J}{9.55 (m M_{dq} - M_{dj})}$$

电动机平均起动转矩: $M_{dq}=79.0\text{N.m}$

Average starting torque of the motor: $M_{dq}=79.0\text{N.m}$

电动机静阻力矩: $M_{dj}=81.4\text{N.m}$ $\left(M_{dj} = \frac{P_{jc} D_{jc}}{2000 i \eta} \right)$

Static resistance torque of the motor: $M_{dj}=81.4\text{N.m}$ $\left(M_{dj} = \frac{P_{jc} D_{jc}}{2000 i \eta} \right)$

折算到电机轴上的总转动惯量: $\sum J=4.8\text{kg.m}^2$ $\sum J = 1.15m(J_1 + J_2) + \frac{(Q + TL + DL)D_0^2}{4i^2\eta}$

Total rotational inertia converted to the motor shaft:

$$\sum J=4.8\text{kg.m}^2 \sum J = 1.15m(J_1 + J_2) + \frac{(Q + TL + DL)D_0^2}{4i^2\eta}$$

起动时间: $t_q=3.1\text{s}$

Starting time: $t_q=3.1\text{s}$

允许的起动时间: $[t_q]=6.0\text{s}$ 满足要求!

Allowable starting time: $[t_q]=6.0\text{s}$ Meet the requirements!

4.2. 起动平均加速度

4.2 Average Starting Acceleration

平均加速度: $a_q=0.30\text{m/s}^2$

Average acceleration: $a_q=0.30\text{m/s}^2$

允许的起动平均加速度: $[a_q]=0.30\text{m/s}^2$ 满足要求!

Allowable average starting acceleration: $[a_q]=0.30\text{m/s}^2$ Meet the requirements!

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5.运行打滑验算（半驱动）

5. Checking Calculation for Operation Slipping (semi-driven)

小车离端梁中心的距离（内侧）： $S_d=1850\text{mm}$

Distance from the trolley from to the center of the end beam (inner side):
 $S_d=1,850\text{mm}$

移动载荷引起的轮压

Wheel pressure caused by moving load

（移动载荷在极限位置）： $P_{Lc61}=40.53\text{kN}$ （满载）

(Moving load at the limit position): $P_{Lc61}=40.53\text{kN}$ (full load)

$P_{Lc62}=1.46\text{kN}$ （空载）

$P_{Lc62}=1.46\text{kN}$ (no-load)

$P_{Lc71}=15.20\text{kN}$ （满载）

$P_{Lc71}=15.20\text{kN}$ (full load)

$P_{Lc72}=0.80\text{kN}$ （空载）

$P_{Lc72}=0.80\text{kN}$ (no-load)

满载最小轮压： $P_{Lcmin}=49.55\text{kN}$

Minimum wheel pressure at full load: $P_{Lcmin}=49.55\text{kN}$

空载最小轮压： $P_{Lcmin}=48.88\text{kN}$

Minimum wheel pressure at no-load: $P_{Lcmin}=48.88\text{kN}$

5.1. 启动

5.1 Starting

$$\left(\frac{\varphi}{K} + \frac{\mu \cdot d}{D}\right) P_{Lmin} \geq \frac{2000i \cdot \eta}{D} \left[M_{dq} - \frac{2000 \times \sum J \times i}{mD} a_q \right]$$

粘着安全系数： $K=1.20$

Adhesion safety coefficient: $K=1.20$

粘着系数： $j=0.120$

Adhesion coefficient: $j=0.120$

$$\left(\frac{\varphi}{K} + \frac{\mu \cdot d}{D}\right) P_{Lmin} = 10.14 \quad \text{kN}$$

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Position: Mechanical Engineer

$$\frac{2000i \cdot \eta}{D} \left[M_{dq} - \frac{2000 \times \sum J \times i}{mD} a_q \right] = 9.85 \quad \text{kN}$$

启动运行不打滑!

No slipping during starting operation!

5.2. 制动

5.2 Braking

$$\left(\frac{\varphi}{K} - \frac{\mu \cdot d}{D} \right) P_{L\min} \geq \frac{2000i}{\eta \cdot D} \left[M_z - \frac{2000 \times \sum J' \times i}{mD} a_z \right]$$

$$\left(\frac{\varphi}{K} - \frac{\mu \cdot d}{D} \right) P_{L\min} = 9.41 \quad \text{kN}$$

$$\frac{2000i}{D \cdot \eta} \left[M_z - \frac{2000 \times \sum J' \times i}{mD} a_z \right] = 7.74 \quad \text{kN}$$

制动运行不打滑!

No slipping during braking operation!

6. 缓冲器

6. Buffer

6.1. 选型

6.1 Model Selection

$$W = \frac{m_{hc} v_c^2}{2} - P_{mt} S = E_t - P_{mt} S = \frac{1}{2} c S^2 \times n$$

最大减加速度: $a_{\max}=4.0\text{m/s}^2$

Maximum deceleration and acceleration: $a_{\max}=4.0\text{m/s}^2$

运行阻力: $P_{mt}=14.07\text{kN}$

Operating resistance: $P_{mt}=14.07\text{kN}$

碰撞瞬时动能: $E_t=13.0\text{kN.m}$

Instantaneous collision kinetic energy: $E_t=13.0\text{kN.m}$

碰撞质量: $m_{hc}=76.6\text{t}$ (满载运行)

Collision mass: $m_{hc}=76.6\text{t}$ (operation at full load)

碰撞前的速度折减系数: 50% (配备限位开关 LX10-12)

Velocity reduction coefficient before collision: 50% (equipped with the limit switch LX10-12)

碰撞速度: $v_c=0.58\text{m/s}$

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Collision velocity: $v_c=0.58\text{m/s}$

单个缓冲器吸收的能量: $W=5.9\text{kJ/kN.m}$

Energy absorbed by single buffer: $W=5.9\text{kJ/kN.m}$

大车缓冲器型号: HCPU100-C

Model of crane buffer: HCPU100-C

缓冲器容量: $[W]=7.455\text{kJ.m}$ 容量满足要求!

Buffer capacity: $[W]=7.455\text{kJ.m}$ The capacity meets the requirements!

缓冲器许用行程: $[S]=79\text{mm}$

Allowable stroke of the buffer: $[S]=79\text{mm}$

最大撞击力: $P_e=380.0\text{kN}$

Maximum impact force: $P_e=380.0\text{kN}$

弹性系数: $c=2.3890\text{kN/mm}$ (弹性缓冲器)

Elastic coefficient: $c=2.3890\text{kN/mm}$ (elastic buffer)

缓冲器效率: $\eta_h=0.85$

Buffer efficiency: $\eta_h=0.85$

单边个数: $n=2.0$

Unilateral number: $n=2.0$

单个重量: $m_{32}=30\text{kg}$

Unit weight: $m_{32}=30\text{kg}$

6.2.缓冲器末端力校核

6.2 Check for End Force of Buffer

缓冲器的末端力:

End force of the buffer:

$$P_{\max} = \frac{cS}{\eta} = 222.04 \quad \text{kN} \quad (\text{弹性缓冲器})$$

$$P_{\max} = \frac{cS}{\eta} = 222.04 \quad \text{kN} \quad (\text{Elastic buffer})$$

缓冲力满足要求!

The buffer force meets the requirements!

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